Statistical Thinking for Data Science

Lecture 6: Bayesian Method



Lecture 6: Bayesian Method

Learning Objectives

- Bayes' theorem and Bayesian analysis
- Bayesian Vs Frequentist statistics
- Prior and posterior distributions
- Bayesian procedure in parameter estimation
- Other Bayesian applications

Lecture 6: Bayesian Method

Bayes' theorem

- Prior information
  - It refers to existing knowledge or beliefs about a particular phenomenon or parameter before new data is collected.
  - Interpreting prior information in Bayesian analysis involves incorporating this prior knowledge into the analysis to update our beliefs based on the new data.

Lecture 6: Bayesian Method

Bayes' theorem

- Prior information example

You are given a six-sided dice and are asked to bet $1 on each roll. If you guess the roll correctly you win $6 otherwise you lose. It stands to reason that this is a game of pure chance, and so your choice of dice roll could be random. However, after a while, you notice that the number 3 comes up a lot more frequently than the others. So much so that you roll a 3 every 5 times out of 6.

You previously assumed that the dice was unbiased, with all numbers equally likely. But you now have information to suggest that rolling a three is much more likely and that the dice is biased. This is an example of existing knowledge.

Lecture 6: Bayesian Method

Bayes' theorem

Prior information example

- How would you use this information to guide how you would bet now?
- This gets at the main idea of Bayesian analysis, where we combine prior information with data to update our expectations.

Lecture 6: Bayesian Method

Prior information

How does this look in a statistical framework?

Suppose you are running a statistical analysis on smoking and cancer risk, there is already information out there examining this research question (previous studies, data, experiments).

Rather than ignoring this prior information, it might be smart to include this as a starting point before you incorporate the new data you have collected.

Prior information

How does this look in a statistical framework?

This means that, if your new data is very surprising—say that there is a negative correlation between smoking and cancer risk—you won't accept this conclusion at face value and state that smoking decreases cancer risk. But you also won't just ignore this evidence.

Lecture 6: Bayesian Method

Bayes' theorem

- The Bayes' theorem describes the probability of an event, based on prior knowledge of conditions that might be related to the event.

$$P(A|B) = \frac{P(B|A) \times P(A)}{P(B)} \quad (1)$$

Where $P(A|B)$ is the probability of event A being true given event B is true.

$P(A|B)$ is the probability of event B being true given event A is true.

$P(A)$ is the independent probability of A being true.

$P(B)$ is the independent probability of B being true.

Lecture 6: Bayesian Method

Bayes' theorem example

- Suppose the probability of a person contracting cancer is 10%, $P(A)$ =0.1.
- This is our prior information, so our best guess for a randomly selected person of contracting cancer would be 10%.
- We then collect additional information about the population, and find that the probability of a person being a smoker is 20%, $P(B)$ =0.2.

Lecture 6: Bayesian Method

Bayes' theorem example

- *New Data:* Additionally, we collect information about cancer patients and find that 80% of cancer patients are also smokers, P(B|A) = 0.8.

- How can we determine the probability of a smoker contracting cancer?

$$Recall\ P(A|B) = P(B|A) * P(A)/P(B)$$

$$P(A|B) = 0.8 * 0.1/0.2 = 0.4$$

Lecture 6: Bayesian Method

Bayes' theorem example

- Our *prior* prediction for a random person sampled from the population having cancer was 0.1.
- However, if we now sample a random person and ask if they were smokers and they said yes, we now expect that they'd have a 0.4 probability of contracting cancer.

Student Survey SFS 2026

Statistical Thinking for Data Science

- Your feedback is very valuable to me as it helps enhance my teaching and recognize effective practices.
- It will only require 2 minutes of your time. I appreciate your participation.
- https://go.blueja.io/dhnxRvxdO0OccJqZp8kDhQ

Thank you!



Lecture 6: Bayesian Method

Questions:

- What is the probability of a randomly selected person who isn't a smoker contracting cancer? Would you expect it to be higher, lower, or the same as the prior probability of 0.1?

- How would you calculate this using the Bayes' theorem equation?

Lecture 6: Bayesian Method

Q1: What is the probability of a randomly selected person who isn't a smoker contracting cancer?

- We know that 80% of cancer patients are also smokers,

  $P(B^C|A) = 1-0.8=0.2$

- How can we determine the probability of a nonsmoker contracting cancer?

  $Recall\ P(A|B^C) = P(B^C|A) * P(A)/P(B^C)$

  $P(A|B^C)=0.2*0.1/0.8=0.025$

Lecture 6: Bayesian Method

Bayes' theorem example

- We now have our updated probabilities that tells us the probability of a randomly selected person contracting cancer if they also smoke.

- What we could then do, is to collect more data, and then update our probabilities again. This could be more data on the rate of cancer in the population P(A), more data on the number of smokers in the population P(B), or further data from known cancer patients who are also smokers P(B|A).

Lecture 6: Bayesian Method

Bayes' theorem example

- We could treat these data as a separate test and then compare the two resulting probabilities for P(A|B), or, we could use this new information to update our previous study results.

- The ability to continually update our conclusions combining prior information with new information is a key idea behind Bayesian analysis.

Lecture 6: Bayesian Method

Bayesian Vs Frequentist statistics

- In lecture one, we've introduced terms like "sample mean" and "population mean", i.e. we've assumed that there is a real value for a population, and we combine data and modelling to try to estimate this parameter.

- For example, we assume that there is a real mean height for the entire population, and we assign a probability of our sample data estimates capturing this "true" parameter.

Lecture 6: Bayesian Method

Bayesian Vs Frequentist statistics

- A Bayesian approach flips this on it’s head a bit. It assumes that the data are “true” and that the parameter, in this case the mean height of the population, has no single value but a range of possible values. But why does this matter?

Lecture 6: Bayesian Method

Bayesian Vs Frequentist statistics

- Thinking about it logically, do you think that there is a definite single population value for the mean height of the population of earth?

- For a given point in time, if we measured the height of everybody, we would find our population parameter, the mean height. But, over time, we may see this population parameter change. For example, the mean height of people increasing over time.

Lecture 6: Bayesian Method

Bayesian Vs Frequentist statistics

Questions:

- For a given point in time, what If we repeated our sampling of heights in the same way each year for 100 years, would we still expect our confidence intervals constructed from data in 1926 to hold up?

- However, our data from 1926 is still relevant to human height, so should we just throw it away? Is this data no longer “true” or useful?

Lecture 6: Bayesian Method

Frequentist confidence intervals

- Let's now extend these ideas from populations and samples to statistical modelling.
- In this course, we've explored linear regression and GLMs, and have examined them from a "frequentist" perspective.

- We've covered confidence intervals and p values, but are based on the idea of repeated, separate experiments. i.e. for a 95% confidence interval, if we repeated this experiment 100 times, we would expect 95 of those confidence intervals to contain our true population parameter.

Lecture 6: Bayesian Method

Bayesian credible intervals

- One of the benefits of a Bayesian framework is that we can construct credible intervals that have an interpretation that matches our intuition.

- Essentially, instead of a single estimate—for example a mean, slope or intercept—with a 95% credible interval, we instead calculate a probability distribution for our parameter.

Lecture 6: Bayesian Method

Bayesian credible intervals

- From this distribution we can then select a range of parameter values that contain 95% of the probability distribution.

- For example, we may have values for average height from 1.52m to 1.9m that capture 95% of the parameter probability distribution. The interpretation of this 95% credible interval is “the probability of the mean height of the population being between 1.52 to 1.90 meters is 0.95”.

Lecture 6: Bayesian Method

Bayesian credible intervals

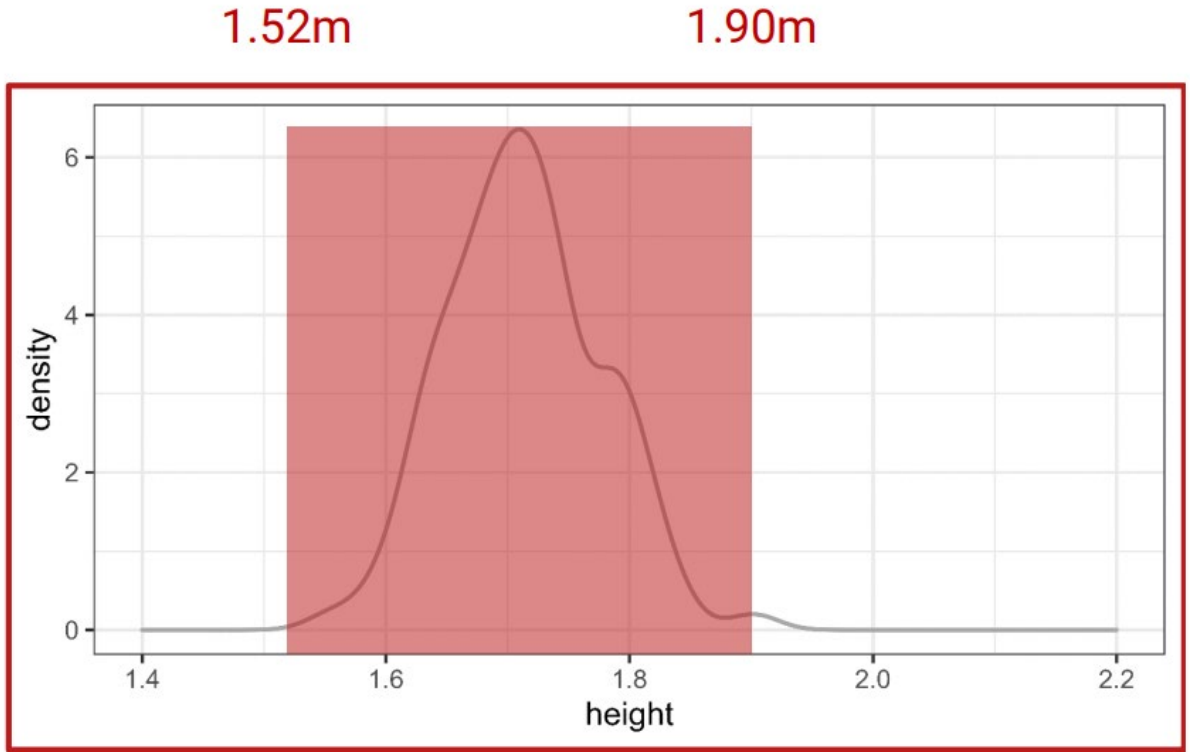


Figure 1: BCI

Summary

Confidence Intervals VS. Bayesian Credible Intervals

Credible intervals capture our current uncertainty in the location of the parameter values and thus can be interpreted as a **probabilistic** statement about the parameter.

Confidence intervals capture the uncertainty about the **interval** we have obtained (i.e., whether it contains the true value or not)

Examples:

95% Confidence intervals of sample height are [1.4 m to 1.8m] : If we repeat this experiment 100 times, we expect 95 of those confidence intervals to contain our true population parameter.

95% Credible intervals of sample height are [1.4 m to 1.8m] : We are 95% confident that the mean height is 1.6m, which is between 1.4 m and 1.8m.

Lecture 6: Bayesian Method

Bayesian Vs Frequentist statistics

- It comes down to how you want to approach decision-making.
- Bayesian 95% credible intervals are easier to interpret but don’t provide you a with single sample parameter easily.
- Frequentist 95% confidence intervals can trip people up in interpretation but do provide a single estimate of the slope, mean, etc, which can be handy to have.

Lecture 6: Bayesian Method

Prior and posterior probability distributions

- As we've covered, a key feature of Bayes' theorem is that is combines prior knowledge and new data to update probabilities.

- But where do you start if you don't actually have any available prior knowledge but still want to utilize a Bayesian approach to construct credible intervals?

Lecture 6: Bayesian Method

Uninformative prior distributions

- Returning to our mean heights problem, if we knew nothing about heights we could start by assigning an equal probability for all height values.

- This is known as an uninformative prior, as there is no actual information about our parameter of interest (mean height) in the probability distribution.

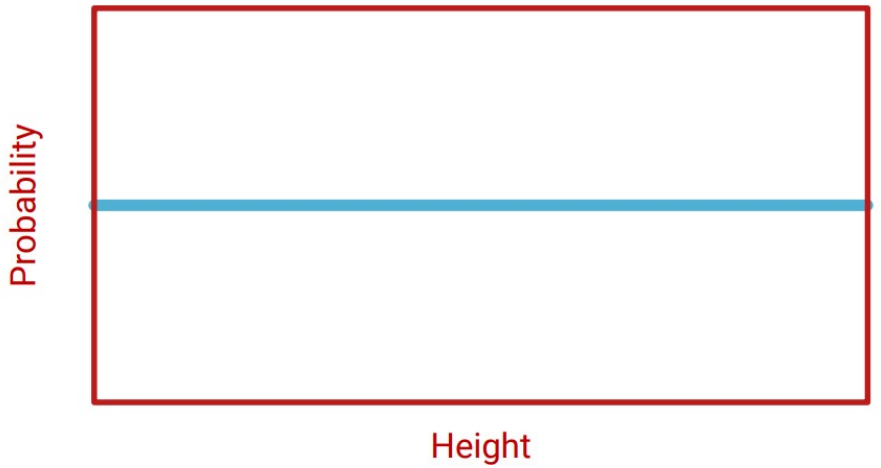
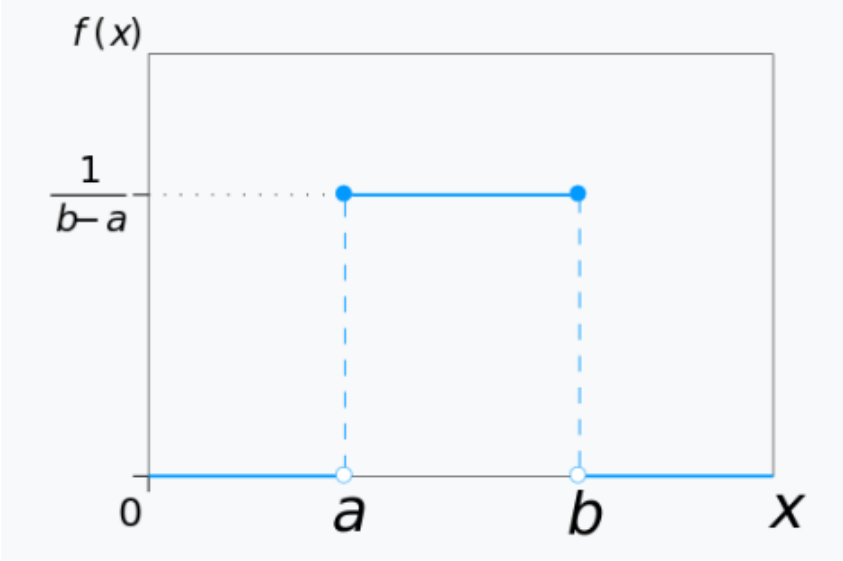


Figure 2: Prior distribution

Lecture 6: Bayesian Method

Uninformative Prior Example

- When you do not have pre-existing information but still want to use Bayesian analysis, a uniform distribution is a common choice.
- Uniform distributions are probability distributions with equally likely outcomes.



$$\theta \sim Uniform\ (a, b)$$

Lecture 6: Bayesian Method

Combining probability distributions: Posterior distributions

- The posterior is the distribution of the parameter, given the data. Bayes' rule :

$$\text{posterior} \propto \text{likelihood} \times \text{prior}.$$

Where $\propto$ means "proportional to".

Lecture 6: Bayesian Method

Combining probability distributions: Posterior distributions

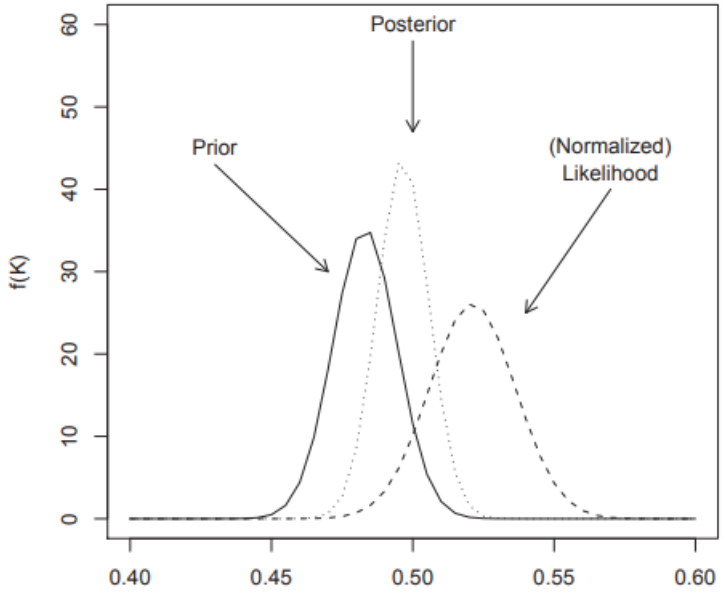


Figure 3: Posterior distribution

Graph taken from bayesbook.pdf.

- The posterior distribution can be written as

$$p(\theta|Y) \propto \mathcal{L}(\theta)\pi(\theta) \quad (2)$$

Where $\mathcal{L}(\theta)$ is the likelihood function and $\pi(\theta)$ is the prior distribution

Lecture 6: Bayesian Method

Combining probability distributions

- We calculate the probability distribution for our new height data, and combine it with the uniform prior distribution via Bayes' theorem to generate our posterior probability distribution.

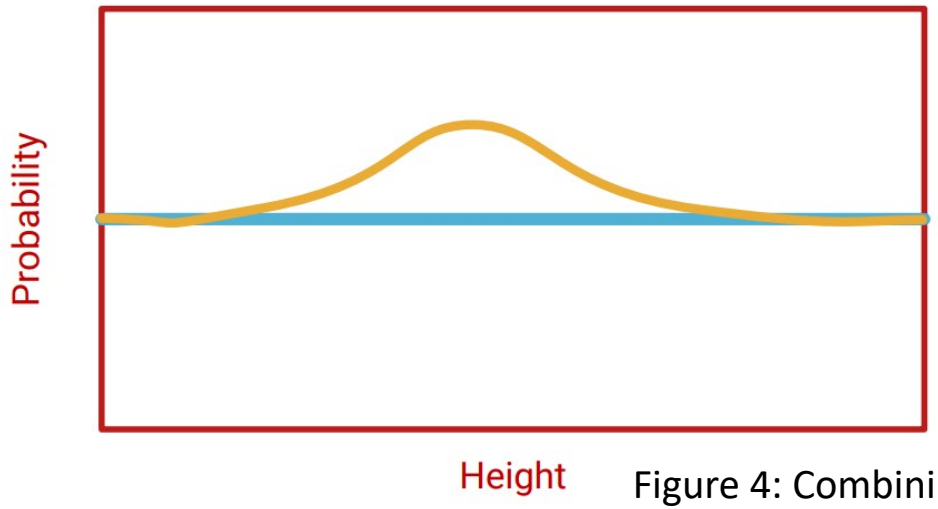


Figure 4: Combining distributions

Lecture 6: Bayesian Method

Posterior distributions with informative prior

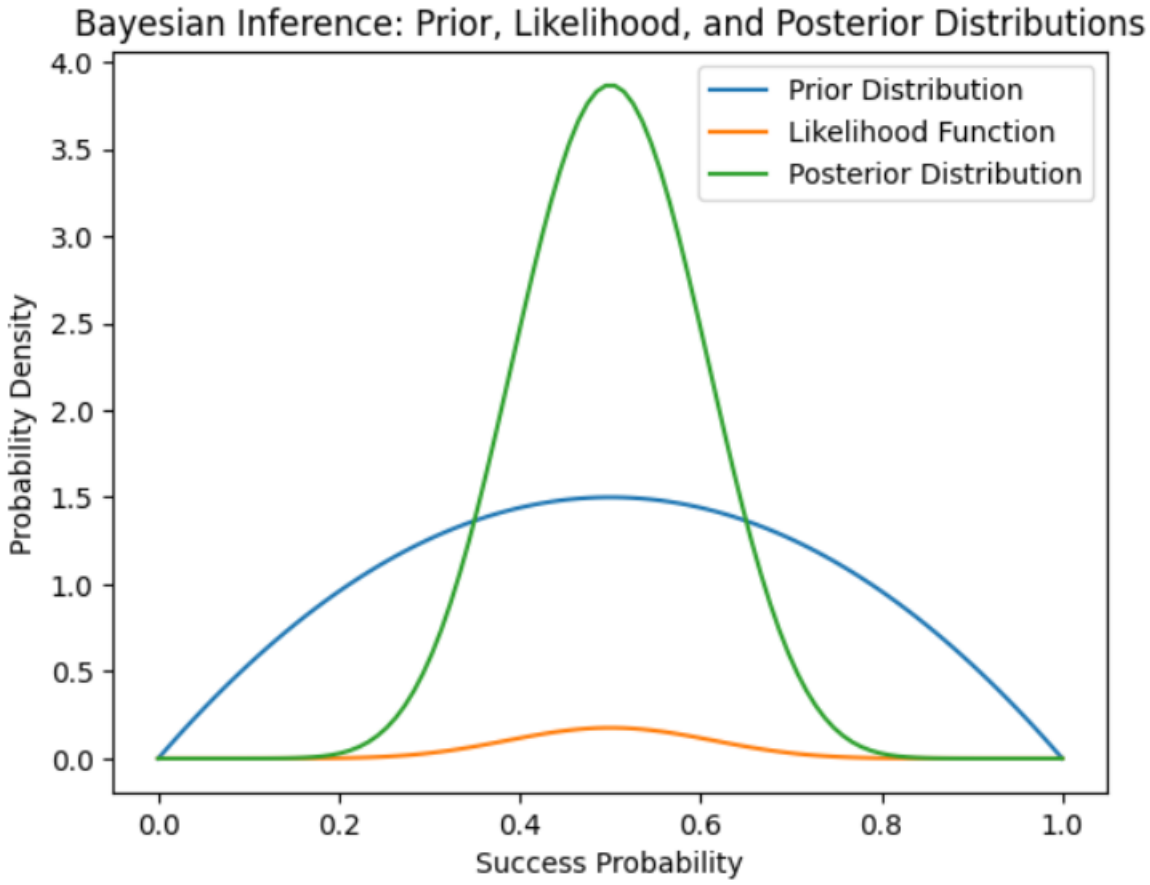


Figure 5: Informative prior

- Suppose we have some prior information about the height (illustrated by the blue curve), we then calculate the probability distribution for our new height data and combine it with the prior distribution via Bayes' theorem to generate our posterior probability distribution (denoted by the green curve).

Lecture 6: Bayesian Method

Bayesian procedure in parameter estimation

- Step 1: We specify the prior distribution $\pi(\theta)$ for each parameter.
- Step 2: We select a statistical model with a likelihood function $\mathcal{L}(\theta)$ that reflects our beliefs about the data y given θ.
- Step 3: After observing data Y, we update our beliefs and calculate the posterior distribution $p(\theta|Y)$.
- Step 4: We can get a Bayesian point estimate by summarizing the center of the posterior.

Lecture 6: Bayesian Method

Example: Applying a Bayesian framework to linear regression

- Below is the output of a Bayesian regression analysis.
- Here the grey lines are different combinations of possible slope and intercept estimates, and the blue line is the average slope and intercept across those possible combinations.

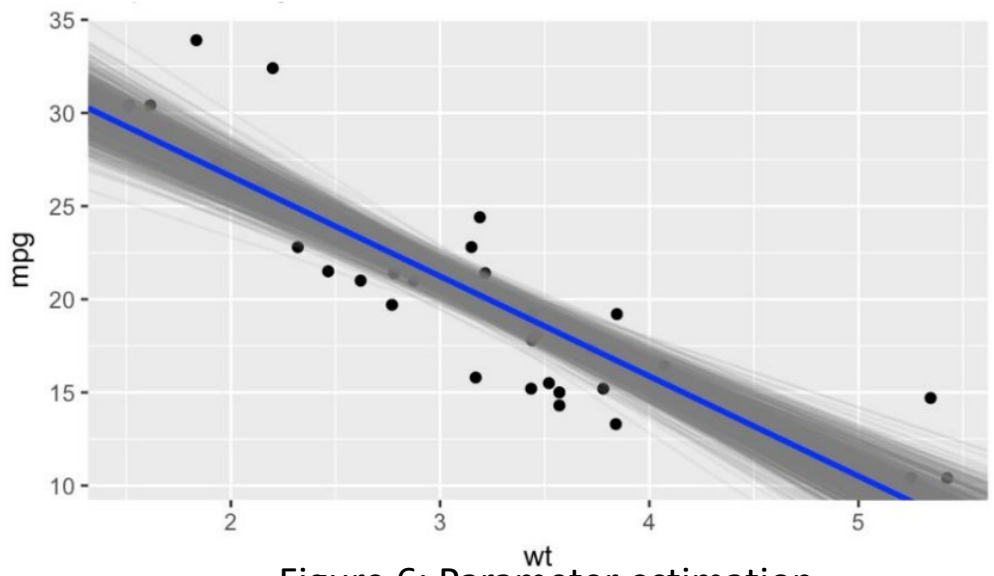


Figure 6: Parameter estimation

Lecture 6: Bayesian Method

Example: Applying a Bayesian framework to linear regression

- Combining probability distributions and finding the prior is expensive work.

- It is worth noting here that calculating the joint conditional posterior probability for multiple parameters (slope, intercept, etc), requires an iterative process.

Lecture 6: Bayesian Method

Markov Chain Monte–Carlo Simulation

- Markov Chain Monte–Carlo (MCMC) is a popular method for obtaining information about distributions, especially for estimating posterior distributions in Bayesian inference.

- This algorithm essentially works to slowly converge on a suitable posterior probability distribution. This is different from the maximum likelihood estimation, and, for large datasets or multiple parameter estimates, can be computationally expensive.

Lecture 6: Bayesian Method

Markov Chain Monte–Carlo Simulation Convergence

Convergence is a critical step in Bayesian analysis to ensure that your MCMC chains have adequately explored the parameter space and have converged to the target distribution.

There are several diagnostics commonly used for assessing MCMC convergence :

- Auto-correlation Plot
- Time Series Plot
- Density Plot

Lecture 6: Bayesian Method

Markov Chain Monte–Carlo Simulation Convergence

Auto-correlation Plots

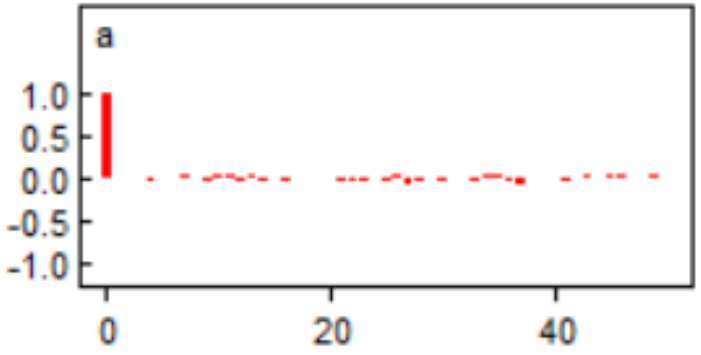
Auto-correlation plots show how each sample is correlated with the previous samples in the chain.

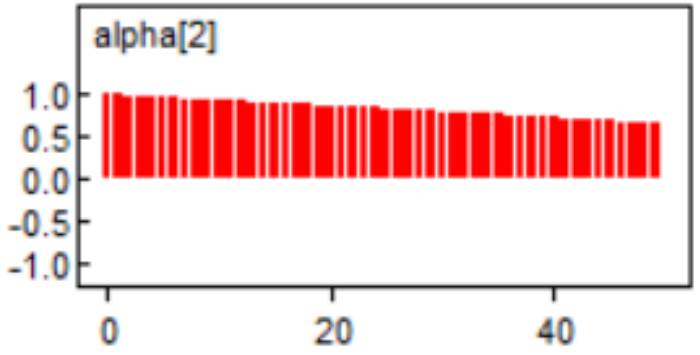
In an ideal MCMC chain, you want the auto-correlation to decrease quickly as the lag increases because this indicates that subsequent samples are less dependent on each other.

Lecture 6: Bayesian Method

Markov Chain Monte–Carlo Simulation Convergence

Auto-correlation measures the correlation between a parameter's value at one step (lag 0) and its value at subsequent steps (lags).

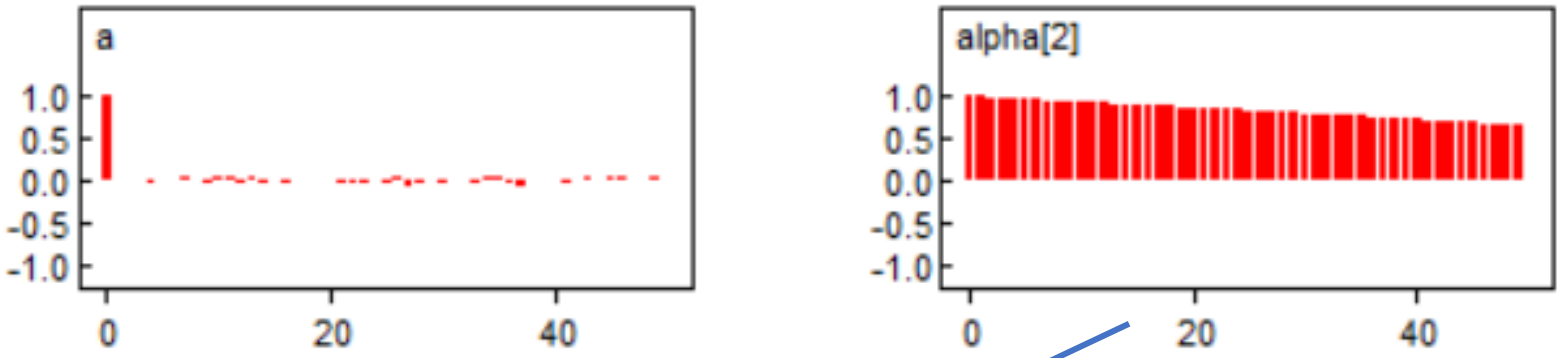




Picture taken from (3)

Lecture 6: Bayesian Method

Auto Correlation



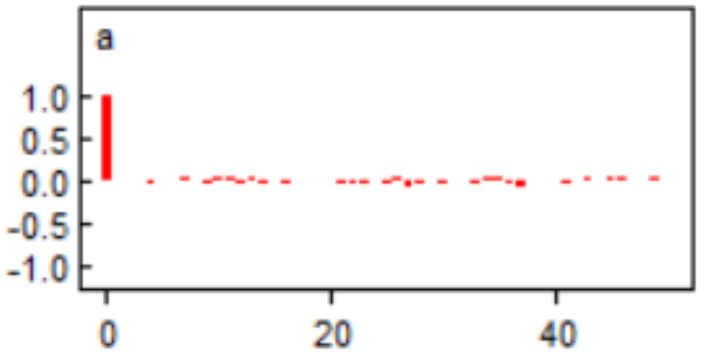
Picture taken from (3)

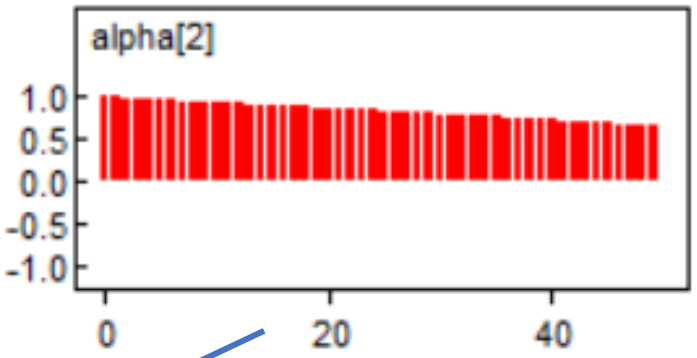
Persistently high autocorrelation

"Thinning" refers to keeping only every kth sample (where k is referred to as the thinning interval). This reduces the autocorrelation by retaining only a subset of samples, making the samples more independent.

Lecture 6: Bayesian Method

Auto Correlation





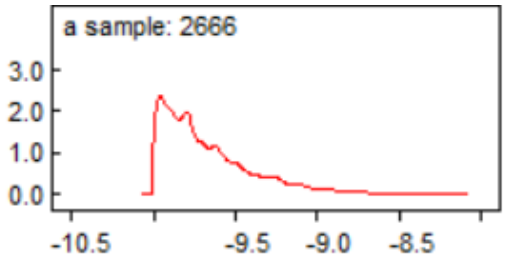
Picture taken from (3)

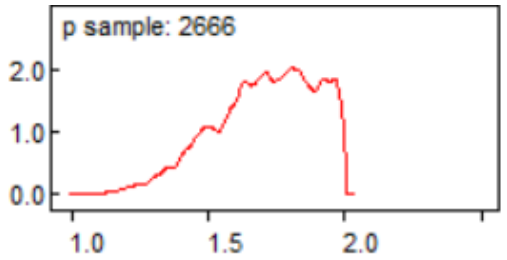
Persistently high autocorrelation

Running the MCMC simulation for a longer period may help reduce autocorrelation.

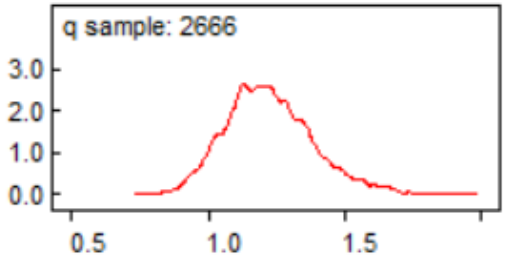
Lecture 6: Bayesian Method

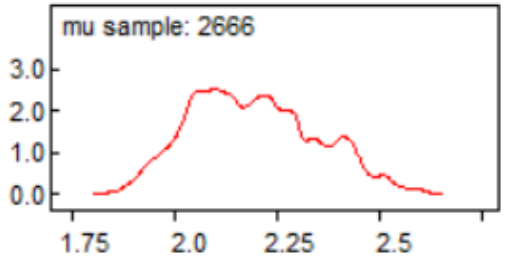
Density Plot





Truncated distribution





Picture taken from (3)

Truncation in the density plot indicates that the sampled values are limited to a subset of the possible parameter values, leading to an incomplete representation of the posterior distribution.

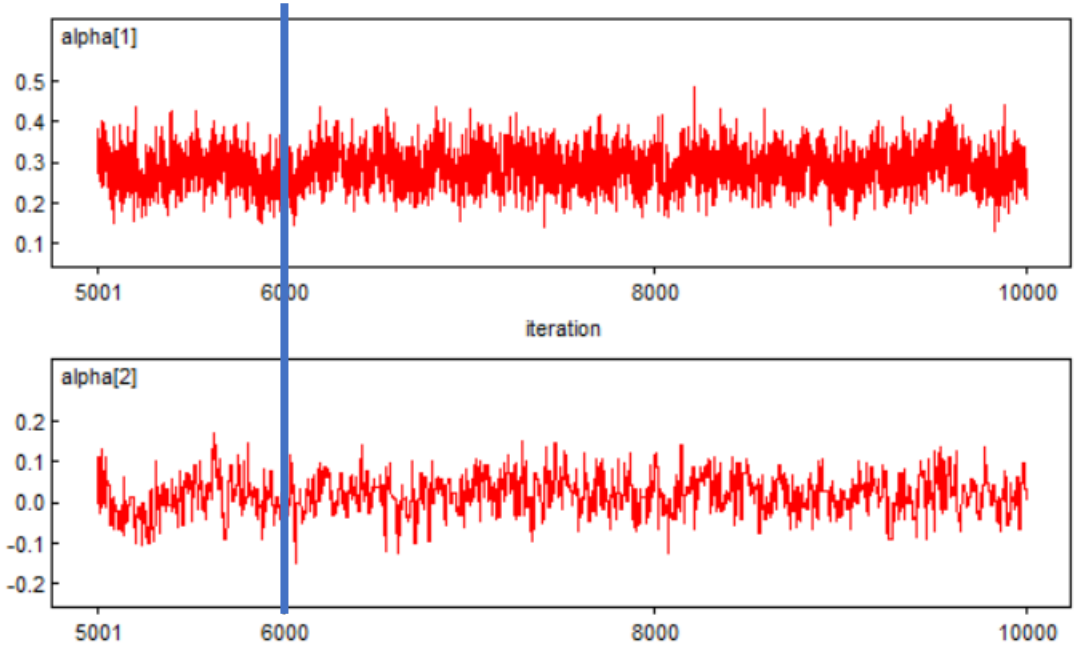
Lecture 6: Bayesian Method

Markov Chain Monte–Carlo Simulation Convergence

Trace Plots

Plotting the MCMC samples as a time series can help identify trends or patterns that may suggest non-convergence.

These plots suggest that the algorithm converged at around iteration 6000.



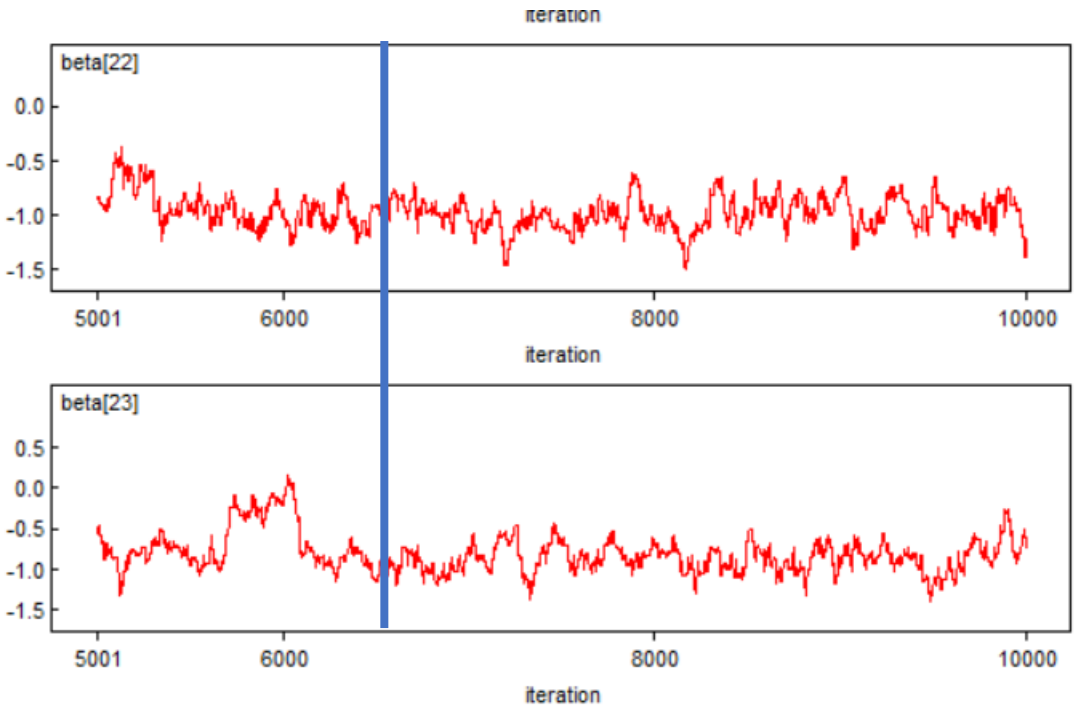
Picture taken from (3)

Lecture 6: Bayesian Method

Markov Chain Monte–Carlo Simulation Convergence

"***burn_in***" is the number of initial samples to discard as burn-in.
You could adjust this value based on your judgment of when the chain has likely converged.

These plots suggest that the algorithm converged at around iteration 6500.



Picture taken from (3)

Lecture 6: Bayesian Method

Bayesian VS. Maximum likelihood Estimation

- So far we have covered MLE and Bayesian. MLE, the frequentist view and Bayesian estimation, the Bayesian perspective, are two prevalent methods employed for parameter estimation.

- MLE estimators have desirable statistical properties. They achieve the smallest possible variance as the sample size increases. However, MLE does not explicitly account for prior information or beliefs, which may be valuable in certain contexts.

Lecture 6: Bayesian Method

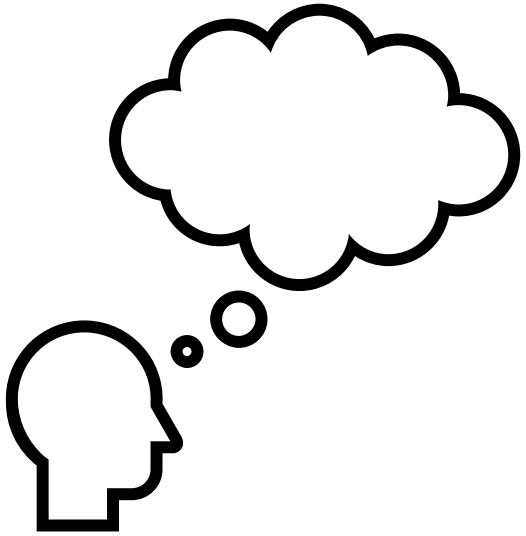
Bayesian VS. Maximum likelihood Estimation

- Bayesian estimation naturally provides measures of uncertainty through posterior distributions, allowing for interval estimates and credible intervals.

- Bayesian methods can handle complex models and datasets, including situations with small sample sizes or high-dimensional problems.

- However, Bayesian estimation can be computationally intensive, especially for complex models or large datasets.

Lecture 6: Bayesian Method



When to use Bayesian?

Lecture 6: Bayesian Method

When to use Bayesian or Maximum likelihood?

We should consider to the following aspects:

- **Is prior Information Available?** Do you have reliable prior knowledge about your problem?

- **How much data do you have?** MLE is often preferred in large-sample scenarios because it is asymptotically efficient, meaning it achieves the smallest possible variance as the sample size increases.

Lecture 6: Bayesian Method

When to use Bayesian or Maximum likelihood?

We should consider to the following aspects:

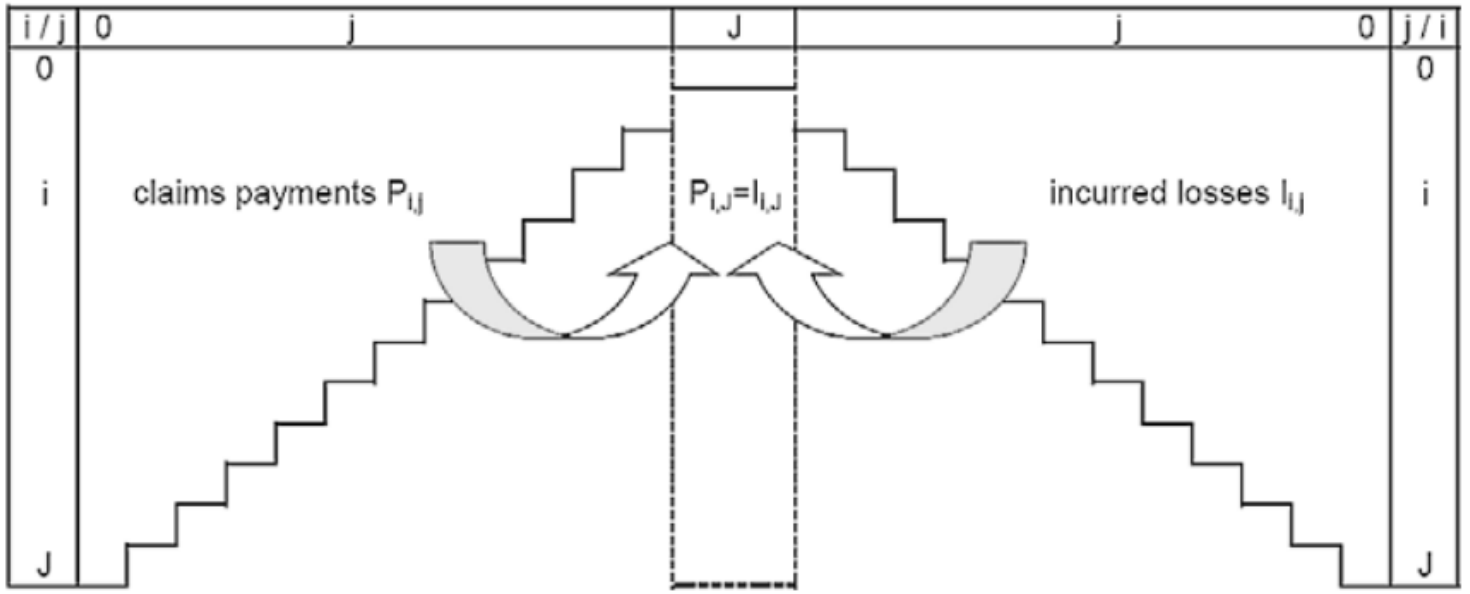
- **Is computational speed critical for you?** Bayesian estimation can involve computational demands, whereas MLE estimation is generally computationally faster in comparison to Bayesian estimation.

- **How complex is your model?** Does it exist a closed-form solution for MLE?

Lecture 6: Bayesian Method

Bayesian Applications

Example: Claims liabilities estimation for long tail class business



Claims Triangle Picture taken from (3)

Lecture 6: Bayesian Method

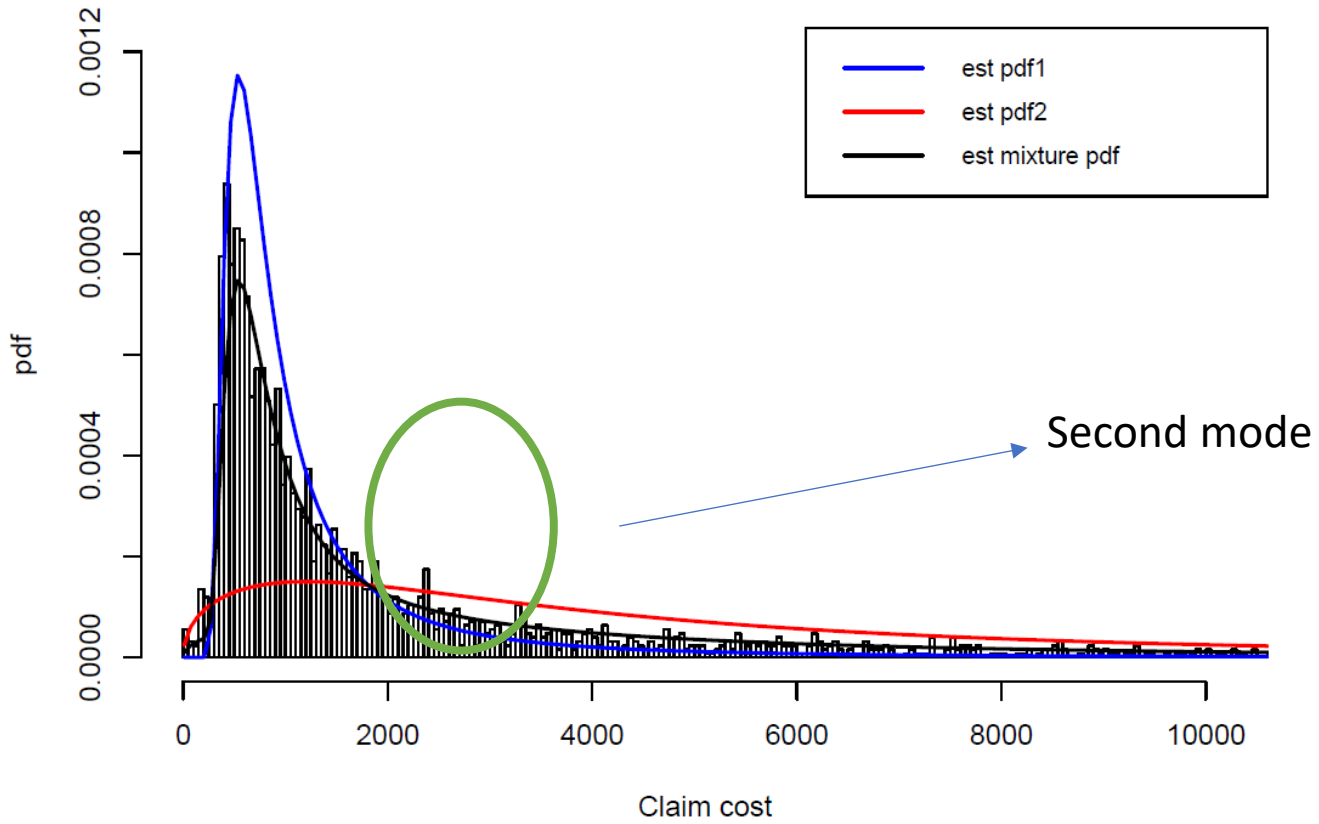
Example: Claims liabilities estimation for long tail class

- A prime example of a long-tail class business in the insurance sector is the Compulsory third-party (CTP) portfolio.
- People who are severely injured in a motor vehicle accident require long-term medical treatment and rehabilitation, resulting in substantially high losses.

Lecture 6: Bayesian Method

Bayesian Applications

Estimating complex models such as mixture models

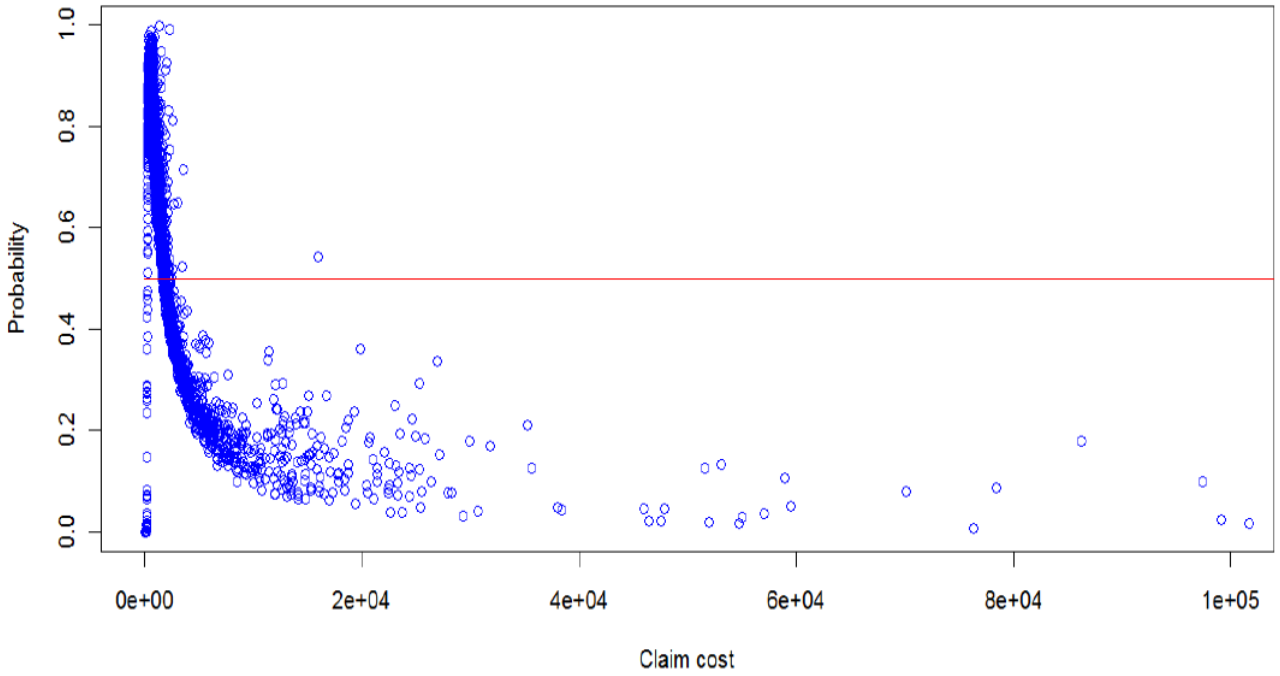


Picture taken from (3)

Lecture 6: Bayesian Method

Bayesian Applications

Estimating complex models such as mixture models



As claims costs increase, the proportion of claims classified to the lower claims group decreases.

Picture taken from (3)

Lecture 6: Bayesian Method

Other Bayesian Applications

- **Bayesian Machine Learning:** Bayesian methods can be used for machine learning tasks such as classification and clustering.

- **Bayesian Neural Networks:** Bayesian neural networks incorporate uncertainty in model parameters and are useful for model uncertainty estimation.

- **Bayesian Optimization:** Bayesian optimization is employed for hyperparameter tuning and optimization tasks.

Lecture 6: Bayesian Software

Most platforms have the capability to execute Bayesian estimations, with varying degrees of efficiency among them.

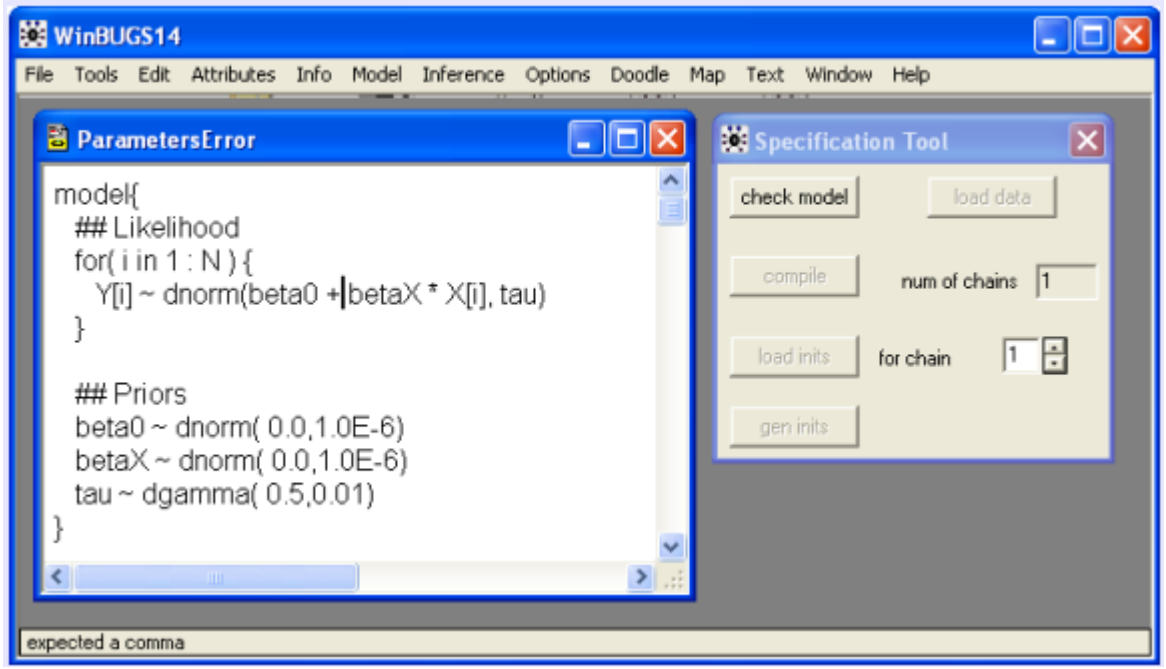
WinBUGS is a free software specifically designed for Bayesian estimation.



Lecture 6: Bayesian Software

WinBUGS has two user interfaces:

- GUI
- Script language



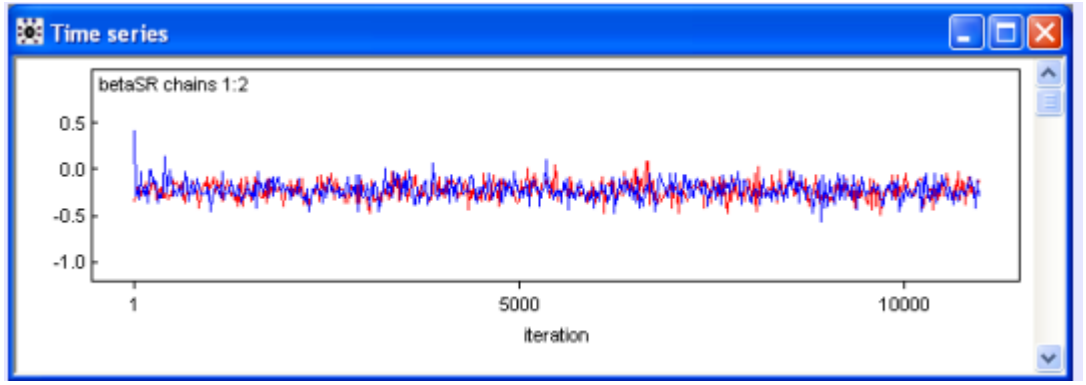
Lecture 6: Bayesian Software

WinBUGS Output and Convergence Diagnostics

- Summary Statistics
- Trace Plot
- Density Plot
- Autocorrelation Plot

Node statistics

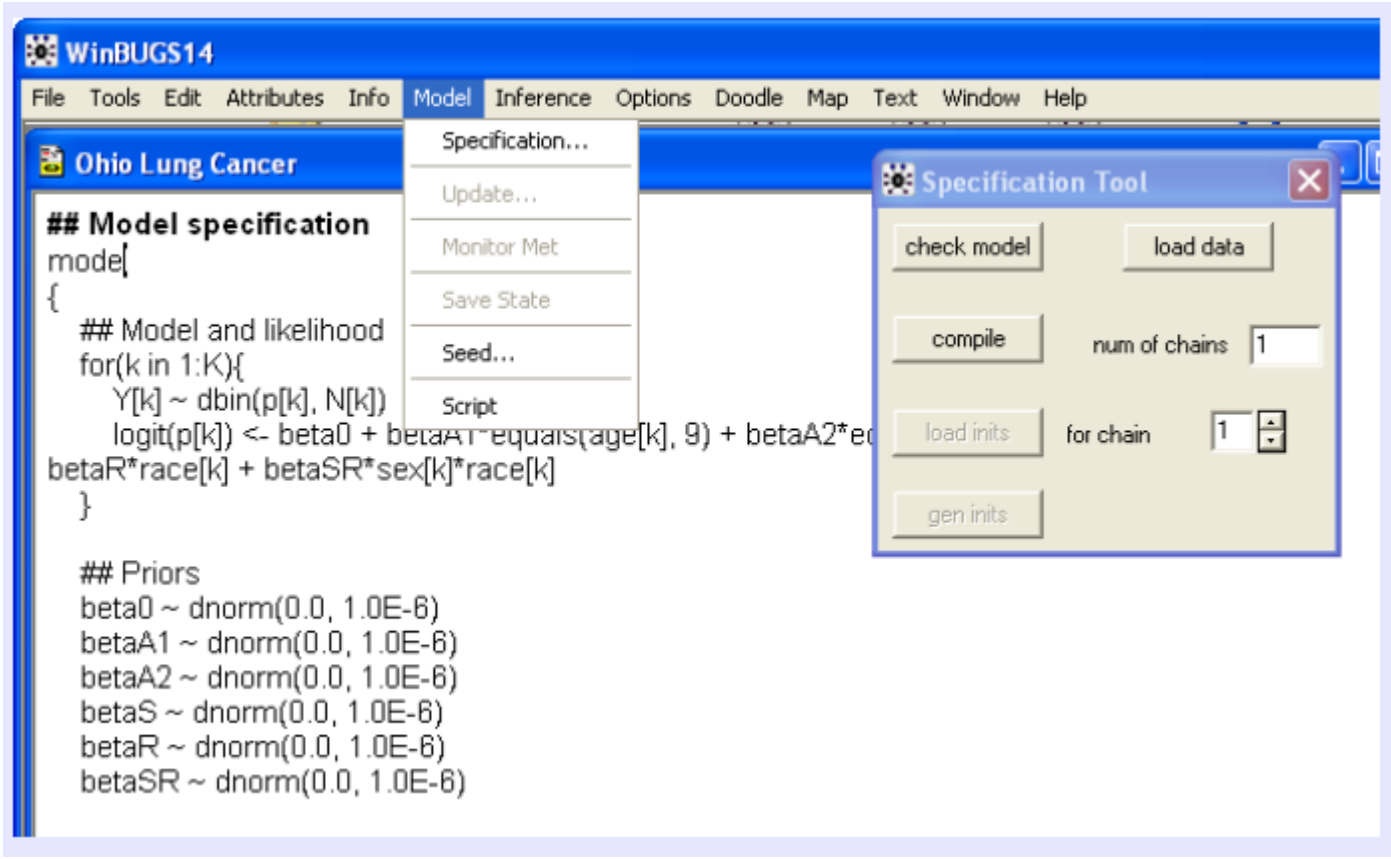
node	mean	sd	MC error	2.5%	median	97.5%	start	sample
beta0	-5.631	0.0267	5.1E-4	-5.686	-5.63	-5.581	1000	20002
betaR	0.3147	0.05165	5.546E-4	0.2137	0.3147	0.4155	1000	20002
betaS	-0.9393	0.03023	3.176E-4	-0.9992	-0.9391	-0.8811	1000	20002
betaSR	-0.2305	0.09374	9.413E-4	-0.4185	-0.2291	-0.04894	1000	20002
sigma	0.1328	0.02858	8.246E-4	0.08105	0.1316	0.1927	1000	20002
theta	0.7976	0.07464	7.446E-4	0.658	0.7953	0.9522	1000	20002



Lecture 6: Bayesian Software

Two Key Pull-down menu lists

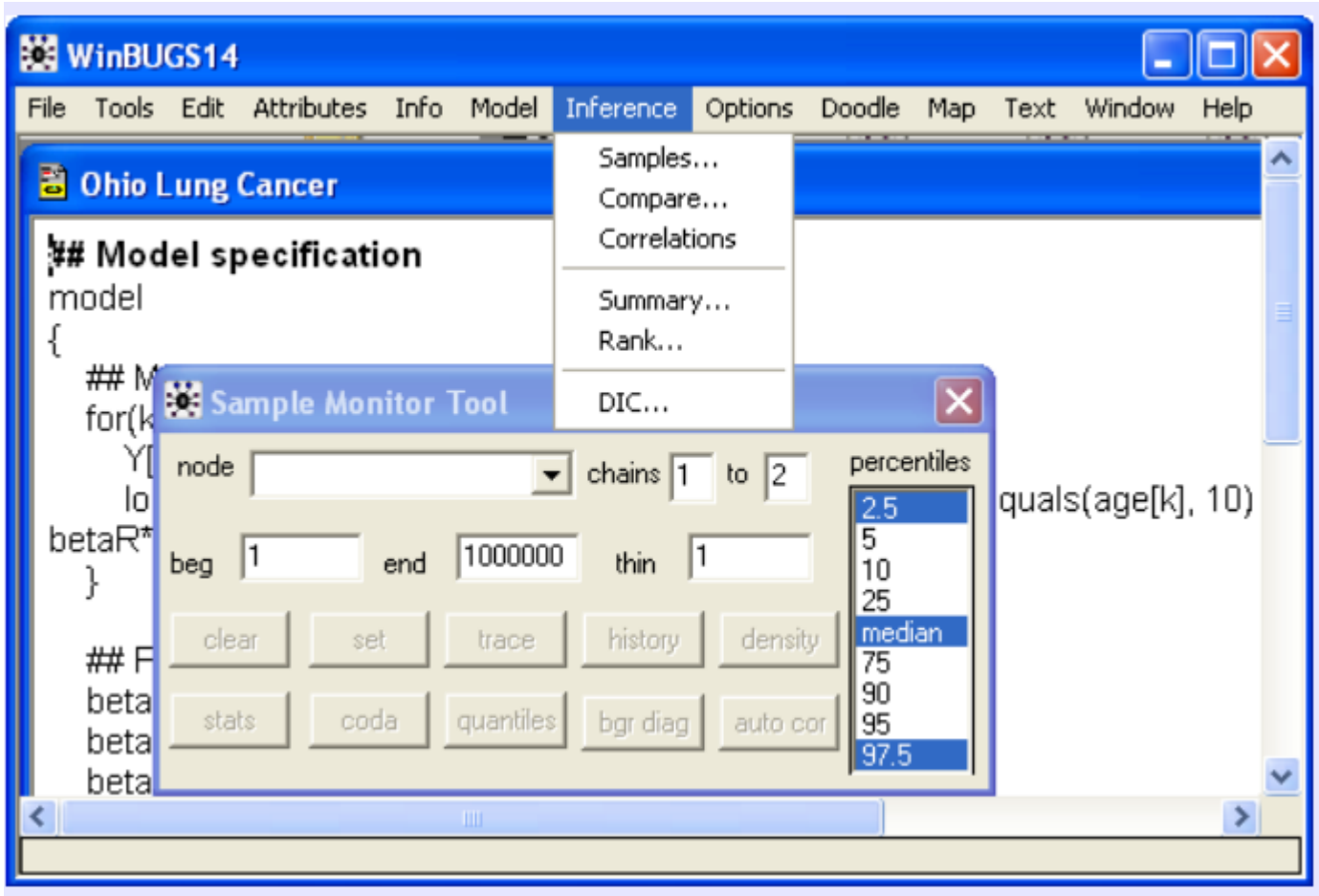
- Model Menu
- Inference Menu



Lecture 6: Bayesian Software

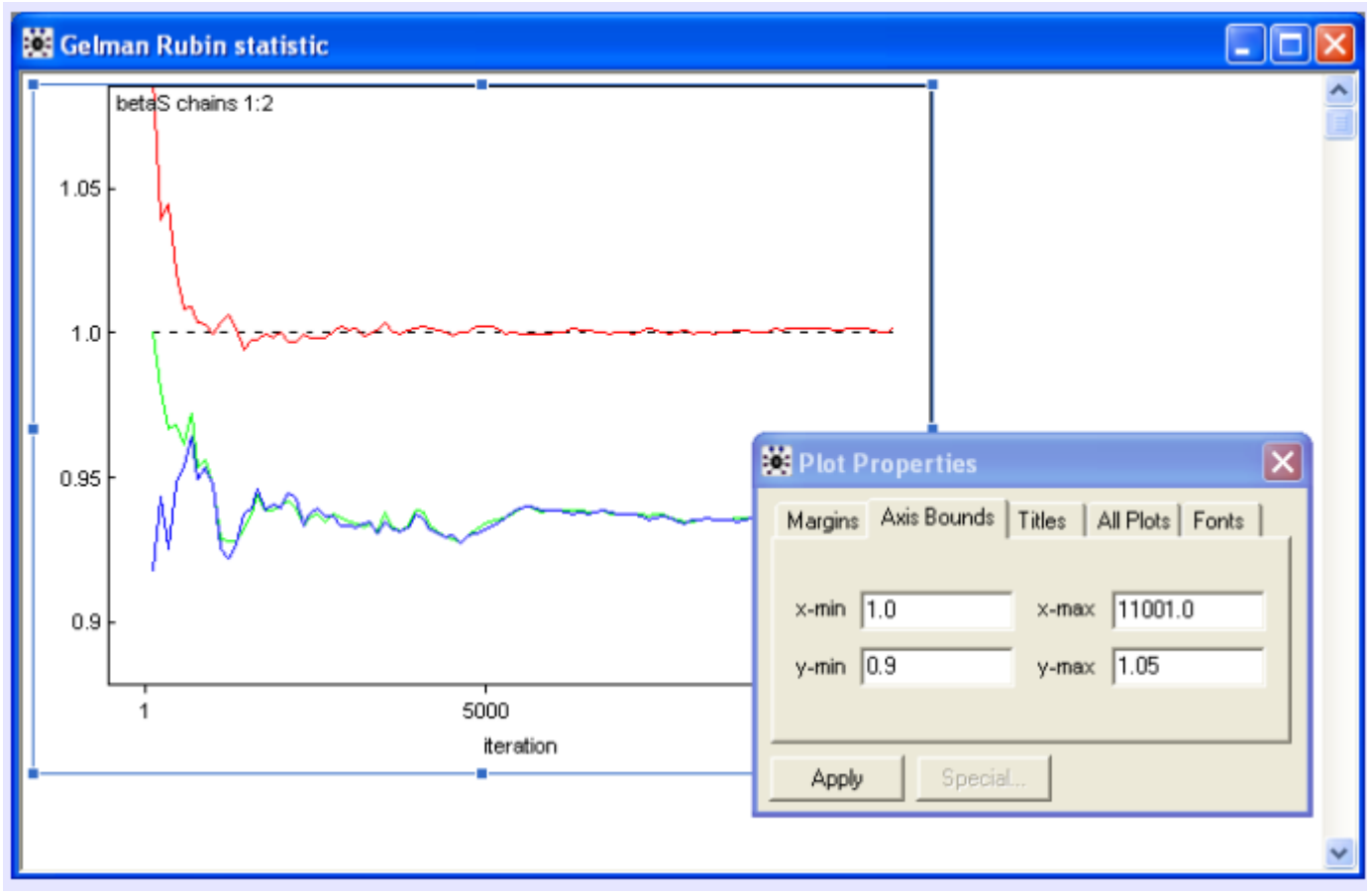
Two Key Pull-down menu lists

- Inference Menu



Lecture 6: Bayesian Software

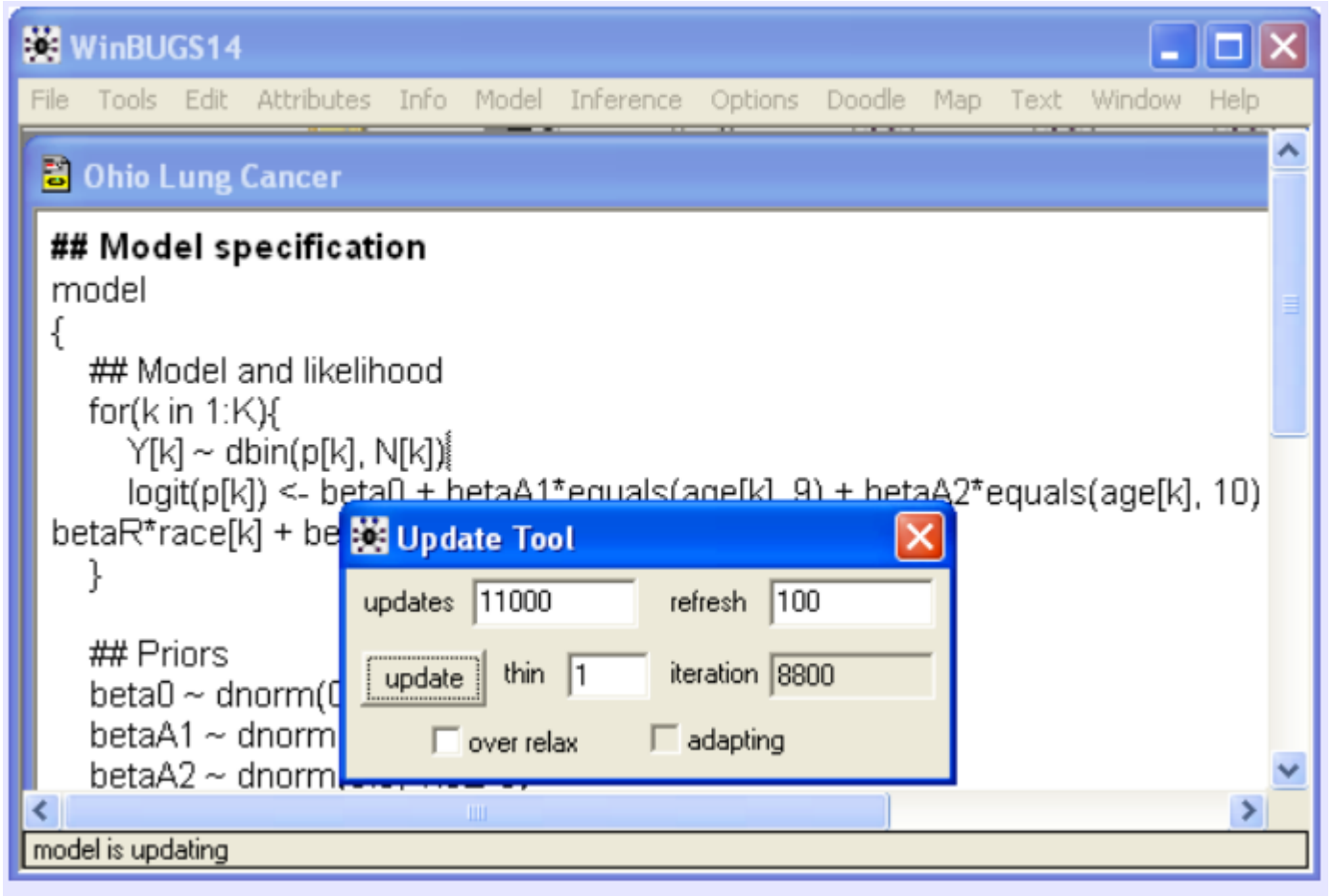
WinBUGS Graphics



Lecture 6: Bayesian Software

WinBUGS Update Tool

- Enter the number of MCMC updates and press ‘update’
- ‘thin’ = k indicates that every k th sample is stored

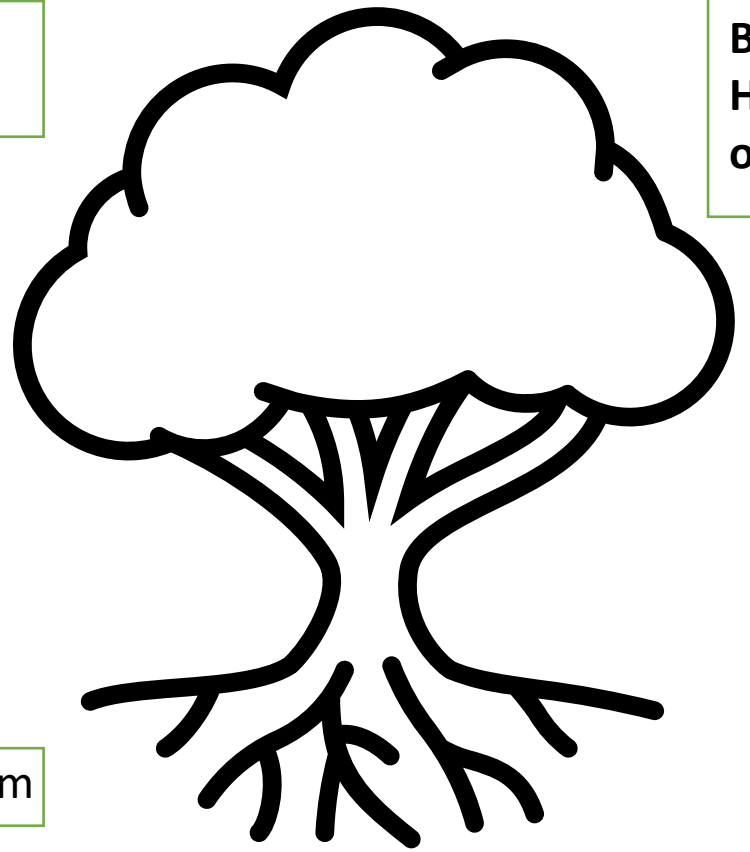


Summary

Applications → **Parameter Estimation**

Bayesian Estimation →

Foundations → Bayes' theorem



Bayesian Neural Network

Bayesian Hyperparameter optimization

Posterior distributions

Prior distributions

Thank you !



If you like my course, please stay in touch. I'd love to hear how things go with your studies or future jobs. If you have any further questions, please feel free to reach out. Xiaodan.dong@uts.edu.au.

References

1. James, G., Witten, D., Hastie, T. and Tibshirani, R. (2021). An Introduction to Statistical Learning with Applications in R (Second Edition). New York: Springer. (An Introduction to Statistical Learning (statlearning.com))

2. Statistical Machine Learning (2022), Chapter 3 , Basics of Bayesian Statistics (bayesbook.pdf)

3. Dong, A. X. D., & Chan, J. S. K. (2013). Bayesian analysis of loss reserving using dynamic models with generalized beta distribution. Insurance, Mathematics & Economics, 53(2), 355–365. https://doi.org/10.1016/j.insmatheco.2013.07.001

4. Bayesian Neural Networks (toronto.edu)

5. Ahamed, S and Uddin, M.M. (2023). Estimating uncertainty of earthquake rupture using Bayesian neural network.

6. Two-dimensional materials-based probabilistic synapses and reconfigurable neurons for measuring inference uncertainty using Bayesian neural networks | Nature Communications

7. Bayesian Optimization: How to calibrate the hyper-parameters of computationally expensive models? | by Jesus Solano | Rappi Tech